

RU final-mine

1 (a)

▼ Define χ^2 variate? Derive the sampling distribution of chi-square (χ^2) variate? **#revise-1**

13.1. Chi-Square Variate (Pronounced as Ki - Sky without S). The square of a standard normal variate is known as a chi-square variate with 1 degree of freedom (d.f.)

Thus if $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$

and $Z^2 = \left(\frac{X - \mu}{\sigma}\right)^2$, is a chi-square variate with 1 d.f.

mgf of chi-square pic

Moment Generating Function (MGF) of Chi-square (χ^2)

Distribution :

We know,
P.d.f of χ^2 is, $f(\chi^2) = \frac{e^{-\frac{\chi^2}{2}} \cdot (\chi^2)^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \cdot \Gamma(\frac{n}{2})}$; $0 < \chi^2 < \infty$

Now, MGF of χ^2 distribution,

$$M_{\chi^2}(t) = E[e^{t\chi^2}] = \int_0^{\infty} e^{t\chi^2} f(\chi^2) d\chi^2$$

Expectation of

$$= \int_0^{\infty} e^{t\chi^2} \frac{e^{-\frac{\chi^2}{2}} \cdot (\chi^2)^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \cdot \Gamma(\frac{n}{2})} d\chi^2$$

$$= \int_0^{\infty} \frac{e^{-\chi^2(\frac{1}{2}-t)}}{2^{\frac{n}{2}} \cdot \Gamma(\frac{n}{2})} d\chi^2$$

[abiz mod nd pristin]

$$\frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \int_0^{\infty} e^{-x^2(\frac{1}{2}-t)} (x^2)^{\frac{n}{2}-1} dx^2 =$$

$$\int_0^{\infty} e^{-ax} x^{\lambda-1} dx = \frac{\Gamma(\lambda)}{a^{\lambda}}$$

$$= \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})} \frac{\Gamma(\frac{n}{2})}{(\frac{1}{2}-t)^{\frac{n}{2}}} =$$

$$= \frac{1}{2^{\frac{n}{2}}} \times \frac{1}{(\frac{1-2t}{2})^{\frac{n}{2}}} = (1-2t)^{-\frac{n}{2}}$$

13.2. Derivation of the Chi-square Distribution.

First Method—Method of Moment Generating Function.

If X_i , ($i = 1, 2, \dots, n$) are independent $N(\mu_i, \sigma_i^2)$, we want the distribution of

$$\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2 = \sum_{i=1}^n U_i^2, \text{ where } U_i = \frac{X_i - \mu_i}{\sigma_i}$$

Since X_i 's are independent, U_i 's are also independent.

$$M_{\chi^2}(t) = M_{\sum U_i^2}(t) = \prod_{i=1}^n M_{U_i^2}(t) = [M_{U_i^2}(t)]^n,$$

since U_i 's $\sim N(0, 1)$ are identically distributed.

Now

$$\begin{aligned} M_{U_i^2}(t) &= E[\exp \{tU_i^2\}] = \int_{-\infty}^{\infty} \exp(tu_i^2) f(x_i) dx_i \\ &= \int_{-\infty}^{\infty} \exp(tu_i^2) \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\} dx_i \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(tu_i^2) \exp(-u_i^2/2) du_i \quad \left[u_i = \frac{x_i - \mu}{\sigma} \right] \end{aligned}$$

13.2

Fundamentals of Mathematical Statistics

$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \left\{ -\left(\frac{1-2t}{2} \right) u_i^2 \right\} du_i \\ &= \frac{1}{\sqrt{2\pi}} \frac{\sqrt{\pi}}{\left(\frac{1-2t}{2} \right)^{1/2}} = (1-2t)^{-1/2} \\ \therefore M_{\chi^2}(t) &= (1-2t)^{-n/2} \quad \dots(13.1a) \end{aligned}$$

1 (b)

▼ Show that χ^2 - distribution tends to normal distribution for large degrees of freedom #revise-1

If $X \sim \chi^2$

then $\chi^2_{(n)} \rightarrow N(n, 2n)$

We know,

M.G.F of χ^2 , $M_X(t) = (1-2t)^{-n/2}$; $|t| < \frac{1}{2}$

The M.G.F of standard χ^2 -variate Z is given by,

$$\begin{aligned} M\left(\frac{X-\mu}{\sigma}\right)(t) &= E\left[e^{t\left(\frac{X-\mu}{\sigma}\right)}\right] \\ &= E\left[e^{\frac{tX}{\sigma} - \frac{t\mu}{\sigma}}\right] \\ &= e^{-\frac{t\mu}{\sigma}} E\left[e^{\frac{tX}{\sigma}}\right] \\ &= e^{-\frac{t\mu}{\sigma}} M_X\left(\frac{t}{\sigma}\right) \\ &= e^{-\frac{t\mu}{\sigma}} \left(1 - 2\frac{t}{\sigma}\right)^{-n/2} \\ &= e^{-\frac{nt}{\sqrt{2n}}} \left(1 - 2\frac{t}{\sqrt{2n}}\right)^{-n/2} \end{aligned}$$

We know,

$$\text{C.G.F, } k_X(t) = \log[M_X(t)]$$

$$\therefore k_Z(t) = \log[M_Z(t)]$$

$$= \log\left[e^{-\frac{nt}{\sqrt{2n}}} \left(1 - 2\frac{t}{\sqrt{2n}}\right)^{-n/2}\right]$$

$$\begin{aligned}
 \Rightarrow K_2(t) &= -\frac{nt}{\sqrt{2n}} - \frac{n}{2} \log \left(1 - 2 \frac{t}{\sqrt{2n}} \right) \quad \left\{ \begin{array}{l} z = \sqrt{2} \sqrt{t} \\ t = \frac{\sqrt{2} \sqrt{t}}{\sqrt{2} \sqrt{n}} = t \sqrt{\frac{2}{n}} \end{array} \right. \\
 &= -\frac{nt}{\sqrt{2n}} - \frac{n}{2} \log \left(1 - t \sqrt{\frac{2}{n}} \right) \\
 &= -t \sqrt{\frac{n}{2}} - \frac{n}{2} \log \left(1 - t \sqrt{\frac{2}{n}} \right) \quad \frac{nt}{\sqrt{2n}} = \frac{\sqrt{n} \sqrt{2} t}{\sqrt{2} \sqrt{n}} = t \sqrt{\frac{n}{2}} \\
 &= -t \sqrt{\frac{n}{2}} - \frac{n}{2} \left[-t \sqrt{\frac{2}{n}} - \frac{1}{2} \left(t \sqrt{\frac{2}{n}} \right)^2 - \frac{1}{3} \left(t \sqrt{\frac{2}{n}} \right)^3 - \dots \right] \\
 &\quad \left[\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \dots \right] \\
 &= -t \sqrt{\frac{n}{2}} + \left[\frac{n}{2} \cdot t \sqrt{\frac{2}{n}} + \frac{n}{2} \cdot \frac{t^2}{2} \left(\frac{2}{n} \right) + \frac{n}{2} \cdot \frac{t^3}{3} \left(\frac{2}{n} \right)^{3/2} - \dots \right] \\
 &= -\cancel{t \sqrt{\frac{n}{2}}} + \cancel{t \sqrt{\frac{n}{2}}} + \frac{t^2}{2} + o_p(n)^{-1/2} \\
 &= \frac{t^2}{2} + o_p(n)^{-1/2}
 \end{aligned}$$

Putting $\lim_{n \rightarrow \infty}$ on both sides,

$$\lim_{n \rightarrow \infty} K_2(t) = \lim_{n \rightarrow \infty} \frac{t^2}{2} + o_p(n)^{-1/2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} K_2(t) = \frac{t^2}{2}$$

$$\Rightarrow M_2(t) = e^{\frac{t^2}{2}} \text{ as } n \rightarrow \infty$$

$\therefore$ Standard χ^2 -variate tends to standard normal variate as $n \rightarrow \infty$. Thus, χ^2 -distribution tends to normal distribution for large d.f.

13.3.2. Limiting Form of χ^2 Distribution for Large Degrees of Freedom. If $X \sim \chi^2_{(n)}$, then $M_X(t) = (1-2t)^{-n/2}$, $|t| < \frac{1}{2}$

The m.g.f. of standard χ^2 -variate Z is given by

$$M_{\frac{X - \mu}{\sigma}}(t) = e^{-\mu/\sigma} M_X(t/\sigma) \quad [\mu = n, \sigma^2 = 2n]$$

$$\begin{aligned} \text{or } M_Z(t) &= e^{-\mu/\sigma} (1 - 2t/\sigma)^{-n/2} \\ &= e^{-n/\sqrt{2n}} \left(1 - \frac{2t}{\sqrt{2n}}\right)^{-n/2} \end{aligned}$$

$$\begin{aligned} \therefore K_Z(t) &= \log M_Z(t) = -t \sqrt{\frac{n}{2}} - \frac{n}{2} \log \left(1 - t \sqrt{\frac{2}{n}}\right) \\ &= -t \sqrt{\frac{n}{2}} + \frac{n}{2} \left[t \cdot \sqrt{\frac{2}{n}} + \frac{t^2}{2} \cdot \frac{2}{n} + \frac{t^3}{3} \left(\frac{2}{n}\right)^{3/2} + \dots \right] \\ &= -t \sqrt{\frac{n}{2}} + t \cdot \sqrt{\frac{n}{2}} + \frac{t^2}{2} + O(n^{-1/2}) \\ &= \frac{t^2}{2} + O(n^{-1/2}), \end{aligned}$$

where $O(n^{-1/2})$ are terms containing $n^{1/2}$ and higher powers of n in the denominator.

$$\therefore \lim_{n \rightarrow \infty} K_Z(t) = \frac{t^2}{2} \Rightarrow M_Z(t) = e^{t^2/2}, \text{ as } n \rightarrow \infty,$$

Exact Sampling Distributions (Chi-square Distribution)

13.7

which is the m.g.f. of a standard normal variate. Hence by uniqueness theorem of m.g.f. Z is asymptotically normal. In other words, standard χ^2 variate tends to standard normal variate as $n \rightarrow \infty$. Thus, χ^2 -distribution tends to normal distribution for large d.f.

1 (c)

- ▼ Write down some important properties and uses of χ^2 - distribution [#revise-1](#)

Properties of Chi-Square Distribution:

1. The shape of the chi-square distribution depends on the degrees of freedom (df).
2. All values of the chi-square random variable are non-negative.
3. It is not bounded
4. It takes values from 0 to positive infinity.

Uses of Chi-Square Distribution:

1. Compares observed data with expected predictions
2. Compares categorical data from different groups
3. Creates confidence intervals for variance
4. Checks if variances are equal between groups

2 (a)

- ▼ Define student's t statistics. Formulate its sampling distribution **#revise-1**

14.2. Student's 't'. Definition. Let $x_i, (i = 1, 2, \dots, n)$ be a random sample of size n from a normal population with mean μ and variance σ^2 . Then Student's t is defined by the statistic

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \quad \dots(14.1)$$

Definition: Let $X_i, i = 1, 2, \dots, n$ be a random sample of size n drawn from a normal population with mean μ and variance σ^2 . Then Student's t-distribution is defined by the statistics

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} \leftarrow$$

Where $\bar{x} = \frac{\sum x_i}{n}$ is the sample mean and $S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$ is the unbiased estimate of the σ^2 .

Derivation of student's t distribution

We know, t statistics,

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$\bar{x} = \frac{\sum x_i}{n}; s = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$\Rightarrow t^2 = \frac{(\bar{x} - \mu)^2}{\left(\frac{s}{\sqrt{n}}\right)^2}$$

$$\Rightarrow t^2 = \frac{(\bar{x} - \mu)^2}{\frac{s^2}{n}}$$

$$\Rightarrow t^2 = \frac{n(\bar{x} - \mu)^2}{s^2}$$

$$\Rightarrow t^2 = \frac{n(\bar{x} - \mu)^2}{\frac{n s^2}{n-1}} \left[s^2 = \frac{n s^2}{n-1} \right]$$

$$\Rightarrow t^2 = \frac{(\bar{x} - \mu)^2}{\frac{s^2}{n-1}} (n-1)$$

$$\Rightarrow \frac{t^2}{n-1} = \frac{(\bar{x} - \mu)^2}{s^2}$$

$$\Rightarrow \frac{t^2}{n} = \frac{\bar{x} - \mu}{s^2} \left[n = n-1 \text{ as d.f.} \right]$$

$$\text{Since } X_i \sim N(\mu, \sigma^2) \Rightarrow \bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$\therefore \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

$$\Rightarrow \left(\frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \right)^2 \sim \chi^2_{(1)}$$

$$\text{i.e., } \frac{n(\bar{x} - \mu)^2}{\sigma^2} \sim \chi^2_{(1)}$$

Now,

$$\frac{t^2}{n} = \frac{n(\bar{x} - \mu)^2}{\sigma^2} \cdot \frac{\sigma^2}{n s^2}$$

$$\frac{n s^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

$$\therefore \frac{t^2}{n} = \frac{\chi^2_{(1)}}{\chi^2_{(n-1)}} \quad \text{--- (1)}$$

Which is the ratio of two χ^2 distribution

Equation follows Beta distribution of second kind.

$$\text{i.e., } \frac{t^2}{v} \sim B\left(\frac{1}{2}, \frac{v}{2}\right)$$

$$\text{We know, Pdf } f(x) = \frac{1}{B(m, n)} \frac{x^{m-1}}{(1+x)^{m+n}}; m, n > 0; x > 0$$

$$\therefore df(t) = \frac{1}{B\left(\frac{1}{2}, \frac{v}{2}\right)} \frac{\left(\frac{t^2}{v}\right)^{\frac{1}{2}-1}}{\left(1+\frac{t^2}{v}\right)^{\frac{v+1}{2}}} d\left(\frac{t^2}{v}\right); 0 \leq t^2 < \infty$$

$$\left(\frac{t^2}{v}\right)^{\frac{1}{2}-1} = \left(\frac{t^2}{v}\right)^{-\frac{1}{2}} = \sqrt{\frac{v}{t^2}} = \frac{\sqrt{v}}{t}; d\left(\frac{t^2}{v}\right) = \frac{2t}{v} dt$$

$$= \frac{2}{\sqrt{v}} dt$$

→ it will disappear bec total range of the probability $(-\infty, \infty)$

$$\Rightarrow df(t) = \frac{1}{\sqrt{v} B\left(\frac{1}{2}, \frac{v}{2}\right)} \frac{1}{\left(1+\frac{t^2}{v}\right)^{\frac{v+1}{2}}} dt; -\infty < t < \infty$$

$$\Rightarrow f(t) = \frac{1}{\sqrt{v} B\left(\frac{1}{2}, \frac{v}{2}\right)} \frac{1}{\left(1+\frac{t^2}{v}\right)^{\frac{v+1}{2}}}; -\infty < t < \infty$$

14.2.1. Derivation of Student's t-distribution. The expression (14.1) can be re-written as

$$t^2 = \frac{n(\bar{x} - \mu)^2}{S^2} = \frac{n(\bar{x} - \mu)^2}{ns^2/(n-1)} \quad [\because ns^2 = (n-1) S^2]$$

$$\Rightarrow \frac{t^2}{(n-1)} = \frac{(\bar{x} - \mu)^2}{\sigma^2/n} \cdot \frac{1}{ns^2/\sigma^2} = \frac{(\bar{x} - \mu)^2/(\sigma^2/n)}{ns^2/\sigma^2}$$

Since x_i , ($i = 1, 2, \dots, n$) is a random sample from the normal population with mean μ and variance σ^2 ,

$$\bar{x} \sim N(\mu, \sigma^2/n) \Rightarrow \frac{(\bar{x} - \mu)}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Hence $\frac{(\bar{x} - \mu)^2}{\sigma^2/n}$, being the square of a standard normal variate is a chi-square variate with 1 d.f.

Also $\frac{ns^2}{\sigma^2}$ is a χ^2 -variate with $(n-1)$ d.f. (c.f. Theorem 13.5).

Further since $\bar{x}$ and s^2 are independently distributed (c.f. Theorem 13.5), $\frac{t^2}{n-1}$, being the ratio of two independent χ^2 -variates with 1 and $(n-1)$ d.f. respectively, is a $\beta_2\left(\frac{1}{2}, \frac{n-1}{2}\right)$ variate and its distribution is given by :

$$dF(t) = \frac{1}{B\left(\frac{1}{2}, \frac{v}{2}\right)} \cdot \frac{(t^2/v)^{\frac{1}{2}-1}}{\left[1 + \frac{t^2}{v}\right]^{(v+1)/2}} d(t^2/v), 0 \leq t^2 < \infty$$

[where $v = (n-1)$]

$$= \frac{1}{\sqrt{v} B\left(\frac{1}{2}, \frac{v}{2}\right)} \cdot \frac{1}{\left[1 + \frac{t^2}{v}\right]^{(v+1)/2}} dt; -\infty < t < \infty$$

the factor 2 disappearing since the integral from $-\infty$ to ∞ must be unity. This is the required probability function as given in (14.2) of Student's t -distribution with $v = (n-1)$ d.f.

2 (b)

▼ Show that student's t distribution reduces to standard normal distribution for large degree of freedom [#revise-1](#)

Student's t distribution reduces to standard normal distⁿ for large degree of freedom

We know, pdf of t distⁿ is,

$$f(t) = \frac{1}{\sqrt{n} B(\frac{n}{2}, \frac{1}{2})} \frac{1}{(1 + \frac{t^2}{n})^{\frac{n+1}{2}}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} f(t) = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} B(\frac{n}{2}, \frac{1}{2})} \left(1 + \frac{t^2}{n}\right)^{-\frac{(n+1)}{2}} \quad \text{--- (i)}$$

Now, $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} B(\frac{n}{2}, \frac{1}{2})} = \lim_{n \rightarrow \infty} \frac{\frac{\Gamma(\frac{n}{2}) \Gamma(\frac{1}{2})}{\Gamma(\frac{n+1}{2})}}{\sqrt{n} \Gamma(\frac{n}{2}) \Gamma(\frac{1}{2})} \quad \left[B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \right]$

$$= \lim_{n \rightarrow \infty} \frac{\Gamma(\frac{n}{2} + \frac{1}{2})}{\sqrt{n} \sqrt{\pi} \Gamma(\frac{n}{2})} \quad \left[\Gamma(\frac{1}{2}) = \sqrt{\pi} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} \sqrt{\pi}} \left(\frac{n}{2}\right)^{\frac{1}{2}} \left[\frac{\Gamma(n+k)}{\Gamma(n)} = n^k \right]$$

$$= \frac{1}{\sqrt{2\pi}} \quad \text{--- (ii)}$$

Also, $\lim_{n \rightarrow \infty} \left(1 + \frac{t^2}{n}\right)^{-\frac{n}{2} - \frac{1}{2}} = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{t^2}{n}\right)^n\right]^{-\frac{1}{2}} \cdot \left(1 + \frac{t^2}{n}\right)^{-\frac{1}{2}}$

$$= \left(e^{t^2}\right)^{-\frac{1}{2}} \cdot 1 \quad \left[\left(1 + \frac{a}{n}\right)^n = e^a, \left(1 + \frac{a}{n}\right)^m = 1\right]$$

$$= e^{-\frac{t^2}{2}} \quad \text{--- (iii)}$$

Putting eqⁿ (ii) and (iii) values in eqⁿ (i),

$$\lim_{n \rightarrow \infty} f(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} \quad \text{which is pdf of standard normal distⁿ}$$

14.2.5. Limiting Form of t-distribution. As $n \rightarrow \infty$, the p.d.f. of t-distribution with n d.f. viz.,

$$f(t) = \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} \left(1 + \frac{t^2}{n}\right)^{-(n+1)/2} \rightarrow \frac{1}{\sqrt{2\pi}} e^{-t^2/2}, \quad -\infty < t < \infty$$

$$\begin{aligned} \text{Proof. } \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \frac{\Gamma[(n+1)/2]}{\Gamma(\frac{1}{2}) \Gamma(n/2)} \\ &= \frac{1}{\sqrt{n}} \cdot \frac{1}{\sqrt{\pi}} \left(\frac{n}{2}\right)^{\frac{1}{2}} = \frac{1}{\sqrt{2\pi}} \end{aligned}$$

$$\left[\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \text{ and } \lim_{n \rightarrow \infty} \frac{\Gamma(n+k)}{\Gamma(n)} = n^k, \text{ (c.f. Remark to § 14.5.7)} \right]$$

Exact Sampling Distributions (t, F and Z Distributions)

14.15

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} f(t) &= \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n} B\left(\frac{1}{2}, \frac{n}{2}\right)} \cdot \lim_{n \rightarrow \infty} \left[\left(1 + \frac{t^2}{n}\right)^n \right]^{-\frac{1}{2}} \\ &\quad \times \lim_{n \rightarrow \infty} \left(1 + \frac{t^2}{n}\right)^{-\frac{1}{2}} \\ &= \frac{1}{\sqrt{2\pi}} \exp(-t^2/2), \quad -\infty < t < \infty \end{aligned}$$

Hence for large d.f. t-distribution tends to standard normal distribution.

2 (c)

▼ Write down some important properties and use of F-distribution **#revise-1**

Properties of F-Distribution:

1. The F-distribution is positively skewed and its shape depends on the degrees of freedom of the numerator and denominator.
2. df1 and df2 are the degrees of freedom of the numerator and denominator, respectively.
3. Values of the random variable are non-negative.

4. For large degrees of freedom, the F-distribution approximates a normal distribution.

Uses of F-Distribution:

1. Used to compare means of three or more groups.
2. In regression, the F-distribution is used to test the overall significance of the regression model.
3. Used to compare variances of two or more groups.
4. used in quality control to compare variations between different processes or treatments.
5. Used to compare nested models in statistical modeling.
6. Applied in clinical trials and medical research for comparing treatment groups.
7. Tests the overall significance of multivariate models.

3 (a)

- ▼ Define point estimator. What are the desirable criteria of good estimation? **#revise-1**

Point estimators are defined as functions that can be used to find the approximate value of a particular point from a given population parameter.

or, a **point estimator** is a single value that is used to estimate an unknown population parameter. For example, if we want to estimate the average height of all students in a school, we could take a sample of students and calculate the average height of the sample. This average height would be a point estimator of the population mean.

Desirable Criteria of Good Estimation:

1. **Unbiasedness:** An estimator is unbiased if its expected value is equal to the true value of the parameter being estimated. In other words, on average, it does not consistently overestimate the parameter.
2. **Efficiency:** An efficient estimator has the smallest possible variance among all estimators for the same parameter. It provides precise and accurate estimates by minimizing the spread of the estimates around the true value.
3. **Consistency:** A consistent estimator becomes increasingly accurate as the sample size increases. It converges in probability to the true parameter value as the sample size approaches infinity.
4. **Sufficiency:** A sufficient estimator contains all the essential information from the sample needed to make unbiased inferences about the parameter. It efficiently captures the relevant information without unnecessary redundancy.

3 (b)

▼ What are the different method of point estimation? **Describe the principles of maximum likelihood method of estimation** #revise-1

15.10. Methods of Estimation. So far we have been discussing the requisites of a good estimator. Now we shall briefly outline some of the important methods for obtaining such estimators. Commonly used methods are

- (i) *Method of Maximum Likelihood Estimation.*
- (ii) *Method of Minimum Variance.*
- (iii) *Method of Moments.*
- (iv) *Method of Least Squares.*
- (v) *Method of Minimum Chi-square*
- (vi) *Method of Inverse Probability.*

The principles of maximum likelihood estimation are:

1. The likelihood function is a function of the parameters of the probability distribution and the observed data. It measures how likely the observed data is under the assumption that the parameters have certain values.
2. The MLE is the set of parameter values that maximize the likelihood function.
3. The MLE is efficient, which means that it has the smallest possible variance among all unbiased estimators of the parameters.
4. The MLE is consistent, which means that it converges to the true value of the parameters as the sample size increases.

3 (c)

▼ Suppose $x_1, x_2, \dots, x_n$ be a random sample from a distribution having the P.D.E of $f(x : n : p)$ = obtain the MLE of P #revise-1

Maximum Likelihood Estimation for Binomial Distribution

Binomial Distribution: $f(x, p) = {}^n C_x p^x q^{n-x}$; $x = 0, 1, 2, \dots, n$

(i) We know that, $f(x, p) = {}^n C_x p^x (1-p)^{n-x}$ [where $q = 1-p$]

1. Find Likelihood function

$$L f(x, p) = {}^n C_x p^x q^{n-x} (1-p)^{n-x}$$

$$\Rightarrow \ln L f(x, p) = \ln {}^n C_x + x \ln p + (n-x) \ln q$$

$$\Rightarrow \ln L f(x, p) = \ln ({}^n C_x p^x (1-p)^{n-x}) \quad [\text{Taking } \ln \text{ both side}]$$

$$\Rightarrow \ln L f(x, p) = \ln \binom{n}{x} p^x (1-p)^{n-x} = \ln \binom{n}{x} + x \ln p + (n-x) \ln (1-p)$$

$$\Rightarrow \ln L f(x, p) = \ln \binom{n}{x} + x \ln p + (n-x) \ln (1-p)$$

3. Taking partial derivatives w.r.t. p $\frac{\partial}{\partial p} \ln L f(x, p) = 0$

$$\frac{\partial \ln L f(x, p)}{\partial p} = 0$$

$$\Rightarrow 0 + x \frac{1}{p} + (n-x) \left[\frac{1}{1-p} \cdot (-1) \right] = 0$$

$$\Rightarrow 0 + \frac{x}{p} - \frac{(n-x)}{1-p} = 0$$

$$\Rightarrow \frac{x}{p} - \frac{(n-x)}{1-p} = 0$$

$$\Rightarrow \frac{x(1-p) - p(n-x)}{p(1-p)} = 0$$

$$\Rightarrow x - xp - pn + px = 0$$

$$\Rightarrow x - np = 0$$

$$\Rightarrow x = np$$

$$\Rightarrow p = \frac{x}{n}$$

$$\therefore \hat{p} = \frac{x}{n}$$

When a sample of 'm' observation is taken. We know that $f(x, p) = \binom{n}{x} p^x (1-p)^{n-x}$

1. Find likelihood function

$$\left[\hat{p} = \frac{x}{n} \right]$$

$$L f(x, p) = \binom{n-x_1}{x_1} p^{x_1} (1-p)^{n-x_1} \cdot \binom{n-x_1-x_2}{x_2} p^{x_2} (1-p)^{n-x_1-x_2} \cdot \dots \cdot \binom{n-x_1-x_2-\dots-x_m}{x_m} p^{x_m} (1-p)^{n-x_1-x_2-\dots-x_m}$$

$$\Rightarrow L f(x, p) = \prod_{i=1}^m \binom{n}{x_i} p^{x_i} (1-p)^{n-x_i}$$

$$\Rightarrow L f(x, p) = \prod_{i=1}^m \binom{n}{x_i} p^{\sum x_i} (1-p)^{mn - \sum x_i}$$

$$\Rightarrow \ln L f(x, p) = \ln \left[\prod_{i=1}^m \binom{n}{x_i} p^{\sum x_i} (1-p)^{mn - \sum x_i} \right]$$

$$\Rightarrow \ln L f(x, p) = \ln \prod_{i=1}^m \binom{n}{x_i} + \ln p^{\sum x_i} + \ln (1-p)^{mn - \sum x_i}$$

$$\Rightarrow \ln L f(x, p) = \ln \prod_{i=1}^m \binom{n}{x_i} + \sum x_i \ln p + (mn - \sum x_i) \ln (1-p)$$

3. Taking Partial derivative w.r.t. 'p'

$$\frac{\partial \ln L f(x, p)}{\partial p} = 0$$

$$\Rightarrow 0 + \sum x_i \frac{1}{p} + (mn - \sum x_i) \frac{1}{1-p} \cdot (-1) = 0$$

$$\Rightarrow \frac{\sum x_i}{p} - \frac{mn - \sum x_i}{1-p} = 0$$

$$\Rightarrow \frac{\sum x_i (1-p) - p(mn - \sum x_i)}{p(1-p)} = 0$$

$$\Rightarrow \frac{\sum x_i - p \sum x_i - p mn + p \sum x_i}{p(1-p)} = 0$$

$$\Rightarrow \sum x_i - p mn = 0$$

$$\Rightarrow p mn = \sum x_i$$

$$\Rightarrow p = \frac{\sum x_i}{mn}$$

$$\Rightarrow p = \frac{\sum x_i}{m} \cdot \frac{1}{n}$$

$$\Rightarrow p = \frac{\sum x_i}{m} \cdot \frac{1}{n}$$

$$\left[\frac{\sum x_i}{m} = \bar{x} \right]$$

$$\Rightarrow p = \frac{\bar{x}}{n}$$

$$\Rightarrow \bar{x} = n p$$

$$\Rightarrow \bar{x} = n p$$

$$\Rightarrow \bar{x} = n p$$

$$\Rightarrow \bar{x} = n p$$

$$\Rightarrow \bar{x} = n p$$

4 (a)

▼ What do you mean by statistical hypothesis? Distinguish between simple and composite hypothesis **#revise-1**

statistical hypothesis is a statement about a population parameter that is being tested. The hypothesis is typically made about the mean, variance, or other characteristic of the population.

There are two types of statistical hypotheses:

- **Simple hypothesis:** A simple hypothesis specifies a single value for the population parameter. For example, the hypothesis that the population mean is equal to 100 is a simple hypothesis.
- **Composite hypothesis:** A composite hypothesis specifies a range of values for the population parameter. For example, the hypothesis that the population mean is greater than 100 is a composite hypothesis.

Criteria	Simple Hypothesis	Composite Hypothesis
Definition	A simple hypothesis specifies a single value for the population parameter.	A composite hypothesis specifies a range of values for the population parameter.
Example	$H_0 : \mu = 100$	$H_1 : \mu \neq 100$
Nature	Specific and particular	General and inclusive
Inference	Typically used for testing specific claims or values.	Often used when testing for a broader range of possibilities.
Complexity	Simpler in nature	Can be more complex due to the range of values considered.

4 (b)

▼ Describe different steps for testing statistical hypothesis **#revise-1**

1. **State the null hypothesis and the alternative hypothesis:** The null hypothesis is the hypothesis that is assumed to be true unless there is strong evidence to the contrary. The alternative hypothesis is the hypothesis that is accepted if the null hypothesis is rejected.
2. **Set the significance level:** The significance level is the probability of rejecting the null hypothesis when it is actually true. The most common significance level is 0.05, which means that there is a 5% chance of rejecting the null hypothesis when it is actually true.
3. **Calculate the test statistic:** The test statistic is a mathematical formula that is used to compare the sample data to the null hypothesis.
4. **Calculate the p-value:** Once we have completed the test statistic, we need to calculate the p value using the data that we have collected.

5. **Determine the critical value:** The critical value is the value of the test statistic that separates the rejection region from the non-rejection region.
6. **Make a decision:** The decision is to either reject the null hypothesis or fail to reject the null hypothesis. The decision is made based on the p-value and the significance level.

Describing the steps below for testing statistical hypothesis,

Step 1 Set up hypothesis and select the level of significance

H_0 : Null hypothesis (No change, No difference)

H_A : Alternate hypothesis (Investigator's belief)

$H_A: \mu > \mu_0 \rightarrow$ Upper tail test

$H_A: \mu < \mu_0 \rightarrow$ Lower tail test

$H_A: \mu \neq \mu_0 \rightarrow$ Two tail test

Step 2 Select the appropriate test statistics

Step 3 Set up decision rule

The decision rule is a statement that tells us under what circumstances to reject the null hypothesis. The decision rule is based on specific values of the test statistic. It determines whether an upper-tailed, lower-tailed or two-tailed test is proposed.

Step 4 Compute the test statistic.

Here we compute the test statistic by substituting the observed sample data into the test statistic identified in Step 2.

Step 5 Conclusion

The final conclusion is made by comparing the test statistic to the decision rule.

4 (c)

▼ A random sample of 10 persons is selected and education in schooling year is recorded as follows: 5, 2, 0, 4, 16, 14, 10, 10, 6, 5. Do you think that the average schooling year of the person in population is 5. [$t_{(0.05, 9)} = 2.26$] #2015_4b #revise-1

Given in the question:

The claim is that the average schooling years of the persons in the population is 5, so the null and alternate hypotheses can be expressed mathematically as follows:

Null Hypothesis (H_0) : $\mu = 5$ (Claim)

Alternate Hypothesis

$$(H_a) : \mu \neq 5$$

This represents a two-tailed hypothesis test.

Number of samples (n) = 10

Sample mean

$$(\bar{x}) = (5 + 2 + 0 + 4 + 16 + 14 + 10 + 10 + 6 + 5)/10 = 72/10 = 7.2$$

Sample standard deviation can be calculated as:

Sample standard deviation

$$(S) = \left(\sqrt{\frac{(5-7.2)^2 + (2-7.2)^2 + (0-7.2)^2 + (4-7.2)^2 + (16-7.2)^2 + (14-7.2)^2 + (10-7.2)^2 + (10-7.2)^2 + (6-7.2)^2 + (5-7.2)^2}{10-1}} \right) = (5.16)$$

For this analysis, we will use a one-sample mean t-distribution test. The test statistic value can be calculated as:

Test statistic

$$(t) = \left(\frac{(\bar{x} - \mu)}{(S/\sqrt{n})} \right) = \frac{(7.2-5)}{(5.16/\sqrt{10})} = 1.35$$

Test statistic (t) = 1.35 (as calculated previously)
Critical value ($t_{(0.05,9)}$) = 2.26 (given in the question)

Since the calculated test statistic (1.35) is less than the critical value (2.26), we fail to reject the null hypothesis.

Hence, we can conclude that the average schooling years of the persons in the population is 5

The degrees of freedom (df) = $n - 1 = 10 - 1 = 9$. From the t-table, we find the corresponding p-value:

$$p - value = 0.21$$

At the significance level ($\alpha = 0.05$), we fail to reject the null hypothesis as the p-value 0.21 is greater than α . Hence, we can conclude that the average schooling years of the persons in the population is 5.

$$H_0 : \mu = 5 \quad H_1 : \mu \neq 5$$

$$t_{obs} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{7.8-5}{2.8/\sqrt{10}} = 2.92$$

5 (a)

▼ Define Type-1 error, Type-2 error, Level of Significance and degrees of freedom **#revise-1**

Type I Error (False Positive):

A Type I error occurs when we reject a true null hypothesis. Type I error is denoted by the symbol α (alpha) and is referred to as the significance level of the test. Lowering the significance level reduces the probability of Type I error but increases the probability of Type II error.

Type II Error (False Negative):

A Type II error occurs when we fail to reject a false null hypothesis. Type II error is denoted by the symbol β (beta) and represents the probability of failing to reject a null hypothesis when it is actually false

Level of Significance (α): The level of significance (α) is the predetermined threshold used to determine whether to reject the null hypothesis.

Degrees of Freedom: Degrees of freedom refer to the number of values in the final calculation of a statistic that are free to vary.

5 (b)

▼ Let a random sample of size n is drawn from a normal population with mean μ and known variance σ^2 . How would you test the hypothesis that mean is equal to μ_0 #revise-1

To test the hypothesis that the mean (μ) is equal to a specific value (μ_0) for a random sample of size n drawn from a normal population with known variance σ^2 , we can use a z-test.

Here are the steps to test the hypothesis:

1. **Formulate Hypotheses:**

- Null Hypothesis (H_0): $\mu = \mu_0$
- Alternative Hypothesis (H_a): $\mu \neq \mu_0$ (Two-tailed test)

2. **Choose Significance Level (α):**

Decide on the level of significance (α) which represents the maximum allowable probability of making a Type-1 error.

3. **Calculate Test Statistic:**

Calculate the z-test statistic using the sample mean ($\bar{x}$), the hypothesized population mean (μ_0), the known population variance (σ^2), and the sample size (n):

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

4. **Find Critical Values:**

- For a two-tailed test, find the critical values in the standard normal distribution corresponding to $\frac{\alpha}{2}$ in the lower tail and $1 - \frac{\alpha}{2}$ in the upper tail.

5. Compare Test Statistic and Critical Values:

Compare the calculated z-test statistic with the critical values. If the test statistic falls beyond the critical values, reject the null hypothesis. If it falls within the critical values, fail to reject the null hypothesis.

6. Make a Decision:

- If the calculated test statistic falls beyond the critical values, reject the null hypothesis.
- If the calculated test statistic falls within the critical values, fail to reject the null hypothesis.

7. Draw a Conclusion:

Based on the decision made in step 6, conclude whether there is enough evidence to support the alternative hypothesis or not.

5 (c)

▼ The average IQ of university female student in Bangladesh is suspected to be more than the average 110 for all students. A random sample of 64 female students yielded a sample average IQ of 115.5 and standard deviation of 20. Can you conclude that the average score of the female students is really more than 110? ($Z_{0.05} = 1.64$) #revise-1

Given that $\mu = 110$
 Sample size (n) = 64 ($n \geq 30$) $\rightarrow$ large sample
 Sample mean ($\bar{x}$) = 115.5
 Sample standard deviation (s) = 20

Null hypothesis (H_0) : The average IQ score of the women in the population is really equals to 110.
 i.e., $H_0 : \mu = \mu_0 = 110$

Alternative hypothesis (H_1) : The average IQ score of the women in the population is really more than 110.
 i.e., $H_1 : \mu > 110$ (One-tailed test)

Level of Significance : $\alpha = 0.05$

Test statistic : $|z| = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{115.5 - 110}{20/\sqrt{64}} = 2.2$

Critical value : $z_{\alpha} = 1.645$

Conclusion : Here, $|z| > z_{\alpha}$, we reject null hypothesis.
 hence, the average IQ score of the women in the population is really more than 110.

6 (a)

▼ What is contingency table? Show that in case of 2x2 contingency table, the test statistic becomes $\chi^2 = \frac{N(ad-bc)^2}{(a+c)(a+b)(c+d)(b+d)}$, $N = a + b + c + d$ #revise-1

Contingency Table:

A **contingency table** is a table that shows the frequency distribution of two or more categorical variables. The variables are usually displayed in rows and columns, and the frequencies are displayed in the cells of the table.

Chi-Square Test for 2x2 Contingency Table

2x2 contingency Table

	B ₁	B ₂	Row Total (RT)
A ₁	a	b	a+b
A ₂	c	d	c+d
Column Total (CT)	a+c	b+d	N = a+b+c+d

We know that,

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

O_i = observed frequency
 E_i = expected frequency

$E(a) = \frac{(a+b)(a+c)}{N}$
 $E(b) = \frac{(a+b)(b+d)}{N}$
 $E(c) = \frac{(c+d)(a+c)}{N}$
 $E(d) = \frac{(c+d)(b+d)}{N}$

RT x CT

$\chi^2 = \frac{(a-b)^2}{\frac{(a+b)(a+c)}{N}} + \frac{(b-d)^2}{\frac{(a+b)(b+d)}{N}} + \frac{(c-a)^2}{\frac{(c+d)(a+c)}{N}} + \frac{(d-b)^2}{\frac{(c+d)(b+d)}{N}}$

$\chi^2 = \frac{N(ad-bc)^2}{(a+c)(a+b)(c+d)(b+d)}$

Now, $\chi^2 = \frac{[a - E(a)]^2}{E(a)} + \frac{[b - E(b)]^2}{E(b)} + \frac{[c - E(c)]^2}{E(c)} + \frac{[d - E(d)]^2}{E(d)}$ (1)

$a - E(a) = a - \frac{(a+b)(a+c)}{N}$

$= \frac{a(a+b+c+d) - (a+b)(a+c)}{N}$

$= \frac{a^2 + ab + ac + ad - a^2 - ac - ba - bc}{N}$

$= \frac{ad - bc}{N}$

d + 0	d	a
b + 0	b	0
b + d + a = n	b + d	a + 0

Similarly we will get,

$$b - E(b) = \frac{ad - bc}{N} \quad c - E(c) = \frac{ad - bc}{N} \quad d - E(d) = \frac{ad - bc}{N}$$

Substituting these values in (1),

$$\chi^2 = \frac{(ad - bc)^2}{N^2} \left[\frac{1}{E(a)} + \frac{1}{E(b)} + \frac{1}{E(c)} + \frac{1}{E(d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad - bc)^2}{N^2} \left[\frac{N}{E(a)} + \frac{N}{E(b)} + \frac{N}{E(c)} + \frac{N}{E(d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad - bc)^2}{N^2} \left[\frac{1}{(a+b)(a+c)} + \frac{1}{(a+b)(b+d)} + \frac{1}{(a+c)(c+d)} + \frac{1}{(b+d)(c+d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad - bc)^2}{N} \left[\frac{a+b+c+d}{(a+b)(a+c)(b+d)} + \frac{a+b+c+d}{(a+c)(c+d)(b+d)} \right]$$

$$\Rightarrow \chi^2 = \frac{(ad - bc)^2}{N} \left[\frac{1}{(a+b)(a+c)(b+d)} + \frac{1}{(a+c)(c+d)(b+d)} \right]$$

$$\Rightarrow \chi^2 = (ad - bc)^2 \frac{a+b+c+d}{(a+b)(a+c)(c+d)(b+d)}$$

$$\Rightarrow \chi^2 = \frac{N (ad - bc)^2}{(a+b)(a+c)(c+d)(b+d)}$$

$$E(a) = \frac{(a+b)(a+c)}{N}$$

$$E(b) = \frac{(a+b)(b+d)}{N}$$

$$E(c) = \frac{(a+c)(c+d)}{N}$$

and $E(d) = \frac{(b+d)(c+d)}{N}$

a	b	$a+b$
c	d	$c+d$
$a+c$	$b+d$	N

$$\therefore \chi^2 = \frac{[a - E(a)]^2}{E(a)} + \frac{[b - E(b)]^2}{E(b)} + \frac{[c - E(c)]^2}{E(c)} + \frac{[d - E(d)]^2}{E(d)} \quad \therefore (*)$$

$$\begin{aligned} a - E(a) &= a - \frac{(a+b)(a+c)}{N} \\ &= \frac{a(a+b+c+d) - (a^2 + ac + ab + bc)}{N} = \frac{ad - bc}{N} \end{aligned}$$

Similarly, we will get

$$b - E(b) = -\frac{ad - bc}{N} = c - E(c); \quad d - E(d) = \frac{ad - bc}{N}$$

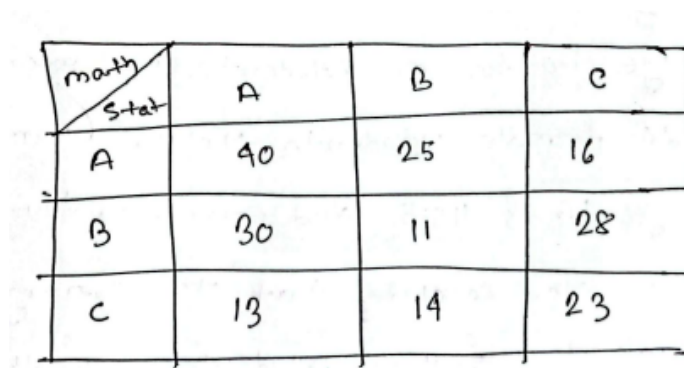
Substituting in (*), we get

$$\begin{aligned} \chi^2 &= \frac{(ad - bc)^2}{N^2} \left[\frac{1}{E(a)} + \frac{1}{E(b)} + \frac{1}{E(c)} + \frac{1}{E(d)} \right] \\ &= \frac{(ad - bc)^2}{N} \left[\left\{ \frac{1}{(a+b)(a+c)} + \frac{1}{(a+b)(b+d)} \right\} \right. \\ &\quad \left. + \left\{ \frac{1}{(a+c)(c+d)} + \frac{1}{(b+d)(c+d)} \right\} \right] \\ &= \frac{(ad - bc)^2}{N} \left[\frac{b+d+a+c}{(a+b)(a+c)(b+d)} + \frac{b+d+a+c}{(a+c)(c+d)(b+d)} \right] \\ &= (ad - bc)^2 \left[\frac{c+d+a+b}{(a+b)(a+c)(b+d)(c+d)} \right] \\ &= \frac{N(ad - bc)^2}{(a+b)(a+c)(b+d)(c+d)} \end{aligned}$$

6 (b)

In the following table below a sample of 200 students have been classified with respect to the grades they received in statistics and mathematics. Is there any association between the grades in the two subjects? **#revise-0**

Math/Stat	A	B	C
A	40	25	16
B	30	11	28
C	13	14	23



math Stat	A	B	C
A	40	25	16
B	30	11	28
C	13	14	23

Test at level of significance 0.05 with tabulated value [$\chi^2_{0.05,4} = 13.3$]

▼ Answer

Let, H_0 : There is no associations betⁿ the grades in the two sub

H_a : " " " " " " " " " " " "

Now, observed Table

Math stat	A	B	C	Row Total (RT)
A	40	25	16	81
B	30	11	28	69
C	13	14	23	50
Column Total (CT)	83	50	67	200

Expected Table

Math stat	A	B	C	Row Total (RT)
A	33.615	20.25	27.135	
B	28.635	17.25	23.115	
C	20.75	12.5	16.75	
Column Total (CT)				

$$E = \frac{RT \times CT}{N}$$

$$E = \frac{RT \times CT}{N}$$

Subject _____

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DATE: / /

O_j	E_j	$(O_j - E_j)^2$	$\frac{(O_j - E_j)^2}{E_j}$
40	33.165	46.71	1.40
30	28.635	1.86	0.06
13	26.75	60.06	2.89
25	20.25	22.56	1.11
11	17.25	39.06	2.26
14	12.5	2.25	0.18
16	27.135	123.98	4.56
28	23.115	23.86	1.03
23	16.75	39.06	2.33

$$\sum \frac{(O_j - E_j)^2}{E_j} = 15.82$$

$$df = (n-1) \times (c-1)$$

$$= (3-1) \times (3-1)$$

$$= 2 \times 2$$

$$= 4$$

Here, $\chi^2_{cal} = 15.82$ and $\chi^2_{tab} = 13.3$

As, $\chi^2_{cal} > \chi^2_{tab}$, we reject the null hypothesis

Hence, there is association between the grades in the two subjects.